

WELL COMPLETION REPORT (SYNTHETIC)

OIL-MRN-010 - Moran-010

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000111
Field / District	Moran / Dibrugarh
Block	PEL-ASSAM-03
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	1998-11-24 / 1999-02-28
Measured / TVD	2804 m / 2770 m
Trajectory	Vertical; inc 4.3 deg; az 36.6 deg
Formation Interval	1559-2567 m
Lithology	Fluvial sandstone with clay streaks
Pore Pressure Reference	3339 psi
Fracture Gradient	15.00 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	High (60.2/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	701	Surface
9 5/8 in casing	1626	Intermediate
7 in liner/casing	2467	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
1999-01-17 1690 m	NPT	Critical	ROP dropped to zero during equipment unavailability. Mitigation: Verify equipment health before resuming drilling. Lesson: Recurring equipment downtime should influence planning and spares.
1998-12-07 526 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Use offset torque trend to set adaptive operating envelope.
1998-12-08 541 m	Kick	High	Unexpected background gas rise preceded influx. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
1998-12-08 557 m	Pressure Spike	High	Pressure trend exceeded recent moving baseline. Mitigation: Condition hole and confirm pressure normalization. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
1998-12-12 666 m	Mud Loss	High	Active pit volume declined during stable pump rate. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
1998-12-14 727 m	NPT	High	No depth progress occurred during the recorded interval. Mitigation: Maintain critical spares for recurring equipment failures. Lesson: Separate geological risk from equipment-driven NPT in analytics.
1999-01-03 1305 m	Kick	High	Flow-out exceeded flow-in and gas units increased. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Escalate connection-gas trends before entering correlated pressure zone.

Completion Summary

The synthetic well reached 2804 m MD in the Tipam interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.